

Phase 1 — Mathematical Modeling

Scope

This document covers Requirements 1 through 5 of the assignment (40% of total grade): formal definition of sets, parameters, decision variables, objective function, hard constraints, and soft constraints for the Invigilator Assignment Problem (IAP) at HCMUT FCSE.

Context and Approach

The Invigilator Assignment Problem requires assigning a fixed number of invigilators to each scheduled exam shift, drawn from a pool of staff, subject to availability rules and fairness preferences. We base our formulation on Bakhtiar et al. (2015) [\[1\]](#), adapted to the HCMUT FCSE setting with three additional preference dimensions:

1. **Travel cost** — invigilators incur a penalty when assigned to shifts away from their home campus.
2. **Fatigue** — same-day consecutive shifts are softly discouraged.
3. **Duration-weighted fairness** — workload is balanced using shift-desirability multipliers, generalizing Bakhtiar's pure-count fairness.

Design decisions

Decision	Rationale
All three roles (CBCT, Thư ký, Trưởng HĐ) treated as one	The optimization concerns invigilation duty assignment; role distinctions are administrative and do not affect spatial/temporal scheduling logic.
Bakhtiar's clash-free constraint omitted	The HCMUT five-slot timetable is non-overlapping; the only possible clash is double-assignment to the same shift, prevented by binary x_{ij} .
Daily cap raised to $k = 4$ (vs. Bakhtiar's $k = 2$)	The HCMUT operating environment routinely schedules more than two shifts per invigilator per day; the value of k is chosen to match the maximum daily load observed during dataset analysis (Phase 2), preserving feasibility while still bounding extreme overloads.
L_1 fairness rather than variance	Variance is quadratic and would force a Mixed-Integer Quadratic Programming (MIQP) formulation. L_1 keeps the model in pure ILP.

Decision	Rationale
Home-campus preference synthesized at 40/40/20 split	Reflects assumed roughly even staff distribution between the two HCMUT campuses (~15 km apart) with a smaller equidistant subset.

Req. 1 — Sets and Parameters

1.1 Indexing sets

Symbol	Definition
E	Set of exam shifts (one per MS Ca thi)
S	Set of invigilators
D	Set of distinct dates in the exam period
C	Set of campuses, $C = \{c_1, c_2\} = \{\text{CS1}, \text{CS2}\}$
$E_d \subseteq E$	Shifts taking place on day $d \in D$
$E_b \subseteq E \times E$	Same-day back-to-back pairs (i, i')

The set E_b is constructed from same-day consecutive time-slot pairs whose inter-slot gap is less than 30 minutes. From the HCMUT five-slot timetable, the qualifying pairs are $(s, s + 1) \in \{(1, 2), (3, 4), (4, 5)\}$. The pair $(2, 3)$ is excluded because of the one-hour lunch gap between slots 2 and 3.

1.2 Shift parameters (extracted from data)

For each shift $i \in E$:

Symbol	Type	Definition
P_i	$\mathbb{Z}_{>0}$	Required number of invigilators
$c(i)$	C	Campus of shift i
$d(i)$	D	Date of shift i
$s(i)$	$\{1, 2, 3, 4, 5\}$	Time slot index
τ_i	$\mathbb{R}_{>0}$	Duration in minutes
μ_i	$\mathbb{R}_{>0}$	Desirability multiplier (defined below)

The desirability multiplier μ_i encodes the relative cost of a shift compared with a baseline weekday afternoon:

$$\mu_i = \mu_{s(i)}^{\text{slot}} \cdot \mu_{d(i)}^{\text{day}} \quad (1.1)$$

with the slot and day multipliers configured as:

$$\mu_s^{\text{slot}} = \begin{cases} 1.1 & s = 1 \quad (07:00 \text{ — early morning}) \\ 1.0 & s \in \{2, 3, 4\} \\ 1.2 & s = 5 \quad (18:15 \text{ — evening}) \end{cases} \quad \mu_d^{\text{day}} = \begin{cases} 1.0 & \text{Mon–Fri} \\ 1.2 & \text{Saturday} \\ 1.3 & \text{Sunday} \end{cases}$$

Justification

Early-morning, evening, and weekend shifts are widely considered less desirable due to commute timing and personal-time impact. Multiplying these slots upward causes them to count more heavily in the workload-fairness goal — an invigilator who absorbs many such shifts contributes proportionally more "weighted workload."

1.3 Synthesized invigilator parameters (simulated location data)

For each invigilator $j \in S$, we synthesize a home-campus assignment $h(j)$ drawn from a fixed categorical distribution:

$$h(j) \sim \text{Categorical}(\{c_1 : 0.4, c_2 : 0.4, \text{eq} : 0.2\}) \quad (1.2)$$

with three semantic categories:

- $h(j) = c_1$ — invigilator resides near Campus 1 (Lý Thường Kiệt, District 10)
- $h(j) = c_2$ — invigilator resides near Campus 2 (Dĩ An, Bình Dương)
- $h(j) = \text{eq}$ — equidistant or no strong preference

Synthesis methodology

The two HCMUT main campuses are approximately 15 km apart. The 40-40-20 split assumes the staff population is roughly evenly divided between the two areas, with a smaller equidistant subset. Synthesis uses Python's `random` module with a fixed seed (`RNG_SEED = 42`) so all team members and reviewers reproduce the same realization.

1.4 Derived parameters

The travel-cost parameter is computed deterministically from h and c :

$$d_{ij} = \begin{cases} 0 & h(j) = c(i) \quad (\text{home match}) \\ 0.5 & h(j) = \mathbf{eq} \\ 1 & h(j) \in C \setminus \{c(i)\} \quad (\text{cross-campus mismatch}) \end{cases} \quad (1.3)$$

The fairness target $\bar{W}$ is the per-invigilator average of total weighted demand:

$$\bar{W} = \frac{1}{|S|} \sum_{i \in E} \mu_i \cdot P_i \quad (1.4)$$

This target is computed once during preprocessing.

1.5 Configuration constants

Symbol	Value	Description
k	4	Hard cap on shifts per invigilator per day
Δ	30 min	Maximum inter-slot gap for back-to-back classification
RNG_SEED	42	Reproducibility of synthesized homes

Req. 2 — Decision Variables

2.1 Primary decision variable

$$x_{ij} = \begin{cases} 1 & \text{if invigilator } j \text{ is assigned to shift } i \\ 0 & \text{otherwise} \end{cases} \quad \forall i \in E, j \in S \quad (2.1)$$

Type: Binary. **Cardinality:** at most $|E| \cdot |S|$ in the dense case.

Variable pruning for scalability

In a sparse implementation, x_{ij} is instantiated only for pairs where invigilator j is potentially eligible for shift i . This is critical when scaling to thousands of shifts and invigilators (Req. 10).

2.2 Auxiliary variable — fatigue

$$y_{ii'j} \in \{0, 1\} \quad \forall (i, i') \in E_b, \forall j \in S \quad (2.2)$$

Type: Binary. **Physical meaning:** $y_{ii'j} = 1$ if and only if invigilator j is assigned to both back-to-back shifts i and i' . This variable linearizes the otherwise bilinear conjunction $x_{ij} \wedge x_{i'j}$ — see constraint (5.2) below.

2.3 Auxiliary variables — workload deviation

$$\alpha_j^-, \alpha_j^+ \in \mathbb{R}_{\geq 0} \quad \forall j \in S \quad (2.3)$$

Type: Continuous, non-negative. **Physical meaning:**

- α_j^- — *under-deviation*: how far below $\bar{W}$ the actual weighted workload of invigilator j falls
- α_j^+ — *over-deviation*: how far above $\bar{W}$ it overshoots

These are deviation variables in the goal-programming sense [1-1]. At optimality, at most one of the two is nonzero per invigilator: any positive value in both could be reduced by the same amount in both, lowering the objective without violating any constraint.

Req. 3 — Objective Function

We minimize a weighted sum of three penalty terms:

$$\min Z = \underbrace{\lambda_1 \sum_{i \in E} \sum_{j \in S} d_{ij} x_{ij}}_{Z_{\text{travel}}} + \underbrace{\lambda_2 \sum_{(i,i') \in E_b} \sum_{j \in S} y_{ii'j}}_{Z_{\text{fatigue}}} + \underbrace{\lambda_3 \sum_{j \in S} (\alpha_j^- + \alpha_j^+)}_{Z_{\text{fairness}}} \quad (3.1)$$

3.1 Term interpretations

Term	Expression	Practical meaning
Travel	$\sum_{i,j} d_{ij} x_{ij}$	Total travel mismatch across all assignments
Fatigue	$\sum_{(i,i'),j} y_{ii'j}$	Number of same-day consecutive double-shifts
Fairness	$\sum_j (\alpha_j^- + \alpha_j^+)$	Total L_1 deviation of weighted workload from $\bar{W}$

3.2 Justification

Why this objective is appropriate. It directly measures three real-world costs that the manual baseline schedule does not optimize for. Travel cost reflects commute time and energy expenditure. Fatigue penalty reflects observed quality degradation in invigilation under back-to-back duty. Fairness deviation prevents any single invigilator from absorbing disproportionate workload — the most common complaint in manual scheduling.

Why L_1 rather than variance. The L_1 form (sum of absolute deviations) keeps the model in pure ILP territory. Variance is quadratic and would require Mixed-Integer Quadratic Programming, restricting solver choice and increasing solution time. The L_1 formulation also tends to spread deviations evenly rather than concentrating them on a few extreme cases.

On the weights $\lambda_1, \lambda_2, \lambda_3$. These are tunable parameters subject to systematic adjustment in Phase 4 (Req. 8). Initial proposed values: $\lambda_1 = 1$, $\lambda_2 = 5$, $\lambda_3 = 1$. The fatigue coefficient is set higher because each occurrence is comparatively rare but particularly objectionable from the staff's perspective.

Req. 4 — Hard Constraints

(H1) Coverage

Every shift must be staffed by exactly its required number of invigilators:

$$\sum_{j \in S} x_{ij} = P_i \quad \forall i \in E \quad (4.1)$$

This rule is non-negotiable: a shift with too few invigilators violates exam-administration policy; with too many, staff resources are wasted.

(H2) Daily cap

No invigilator is assigned more than k shifts on any single day:

$$\sum_{i \in E_d} x_{ij} \leq k \quad \forall d \in D, \forall j \in S \quad (4.2)$$

with k chosen to bound extreme overloads while remaining feasible against the operating environment's actual daily-load patterns. The specific value used in this study is justified in the Design decisions table above and confirmed against the data in Phase 2.

Removed: clash-free constraint

Bakhtiar's formulation includes $\sum_{i \in E_o} x_{ij} \leq 1$ for overlapping time slots. In our HCMUT-adapted model this is structurally redundant: the five fixed time slots are non-overlapping, so the only possible "clash" is double-assignment to the same shift — already prevented by $x_{ij} \in \{0, 1\}$.

(H3) Pre-known unavailability

Implemented at the variable-creation level: x_{ij} is not instantiated for pairs (i, j) where invigilator j is unavailable for shift i . This is more efficient than adding $x_{ij} = 0$ as an explicit linear constraint.

Req. 5 — Soft Constraints

The soft constraints take two forms: **goal constraints with deviation variables** (the goal-programming pattern from Bakhtiar et al.) and **linearization auxiliaries** for non-linear expressions. Each one contributes a corresponding penalty term to the objective.

(S1) Travel penalty

The travel cost d_{ij} is a precomputed parameter, so no algebraic constraint is required. The penalty appears directly as the first term of the objective (3.1):

$$Z_{\text{travel}} = \sum_{i \in E} \sum_{j \in S} d_{ij} x_{ij} \quad (5.1)$$

This is the cleanest possible formulation — campus mismatch is an attribute of the assignment itself, not a state of the model.

(S2) Fatigue — linearized conjunction

For every back-to-back pair and every invigilator:

$$y_{ii'j} \geq x_{ij} + x_{i'j} - 1 \quad \forall (i, i') \in E_b, \forall j \in S \quad (5.2)$$

Linearization argument. In a minimization problem with $y_{ii'j}$ appearing as a positive-coefficient term in the objective, the solver pushes $y_{ii'j}$ toward zero. Constraint (5.2) forces $y_{ii'j} \geq 1$ exactly when both $x_{ij} = 1$ and $x_{i'j} = 1$. In all other cases the right-hand side is non-positive, allowing $y_{ii'j} = 0$. This achieves the AND semantics $y_{ii'j} = x_{ij} \wedge x_{i'j}$ without any bilinear terms, keeping the model in pure ILP territory.

(S3) Duration-weighted workload fairness

For each invigilator, the actual weighted workload should match the target $\bar{W}$:

$$\sum_{i \in E} \mu_i \cdot x_{ij} + \alpha_j^- - \alpha_j^+ = \bar{W} \quad \forall j \in S \quad (5.3)$$

Goal-programming interpretation. Constraint (5.3) is always satisfiable because the deviation variables α_j^-, α_j^+ absorb any discrepancy between the actual weighted workload $\sum_i \mu_i x_{ij}$ and the target $\bar{W}$. The objective then minimizes $\alpha_j^- + \alpha_j^+$, driving each invigilator's workload toward $\bar{W}$ in L_1 distance. If the actual is below the target, $\alpha_j^- > 0$; if above, $\alpha_j^+ > 0$; at most one is positive at optimum.

Why τ_i does not appear in (5.3)

Bakhtiar's general formulation uses $w_i = \tau_i \cdot \mu_i$ where τ_i is shift duration. In the HCMUT dataset, all shifts are 150 minutes — so τ_i is a global constant that cancels out of the goal

equation. The simplified weight $w_i = \mu_i$ retains the meaningful structure (slot/day desirability) without redundant scaling.

1. Bakhtiar, T., Hanum, F., & Romliyah, A. (2015). Exam Invigilators Assignment Problem: A Goal Programming Approach. *Applied Mathematical Sciences*, Vol. 9, No. 58, pp. 2871–2880. ↩ ↩